Learning Spatial Unit-Norm Latents with Riemannian Flow Matching

SRUL: Spherical Riemannian Unit-Norm Latents

Motivation

An autoencoder does not have to produce Gaussian latents

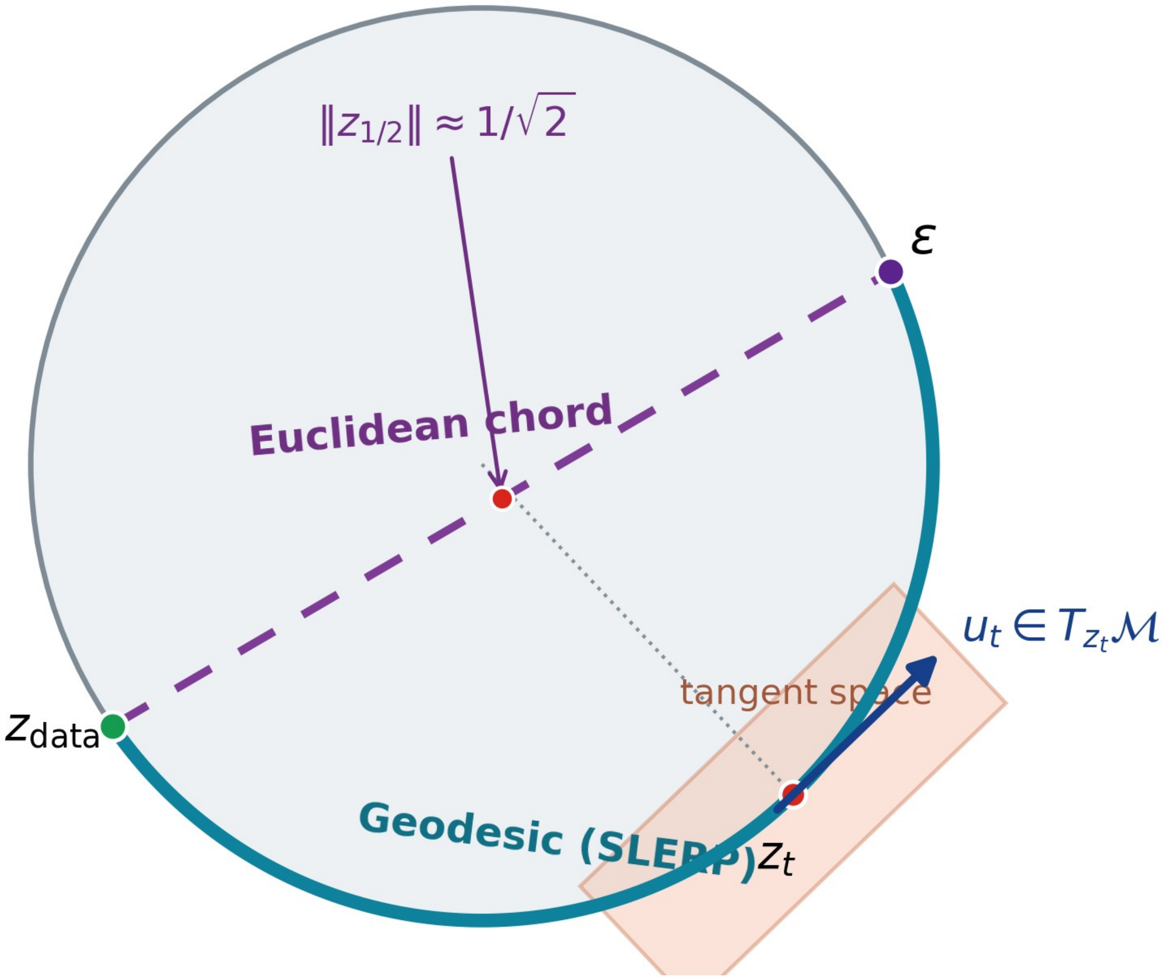
Normalization can make token norms nearly constant
(e.g., Transformer + LayerNorm)

The prior can mismatch the latent support or leave it along the transport path

Main contribution

SRUL learns spherical spatial tokens and uses tangent noise for information control. RFM models the same latent support.

CIFAR-10 • PathMNIST • CelebA-64



SRUL pipeline

Learn spherical latents, control retained information, then sample on the same manifold

1 Learn the spherical representation

Image → Encoder → Unit-norm spatial tokens → Decoder

2 Train the Riemannian Flow Matching prior

Spherical noise → SLERP state z_t → RFM field $v\theta(z_t,t)$ → Tangent velocity → Target u_t

3 Sample new latents and decode

Noise → Projected velocity + Exp-map steps → Latent → Decoder → Image

Unit-norm token

$$z_{i,j} = \frac{h_{i,j}}{\|h_{i,j}\|_2} \in \mathbb{S}^{C-1}$$

Geodesic path

$$z_t = \frac{\sin((1-t)\Omega)}{\sin\Omega} z + \frac{\sin(t\Omega)}{\sin\Omega} \varepsilon$$

Velocity matching

$$\mathcal{L}_{\mathrm{RFM}} = \mathbb{E}\, \|\Pi_{z_t}(v_\theta) - u_t\|_2^2$$

Manifold-preserving sampling

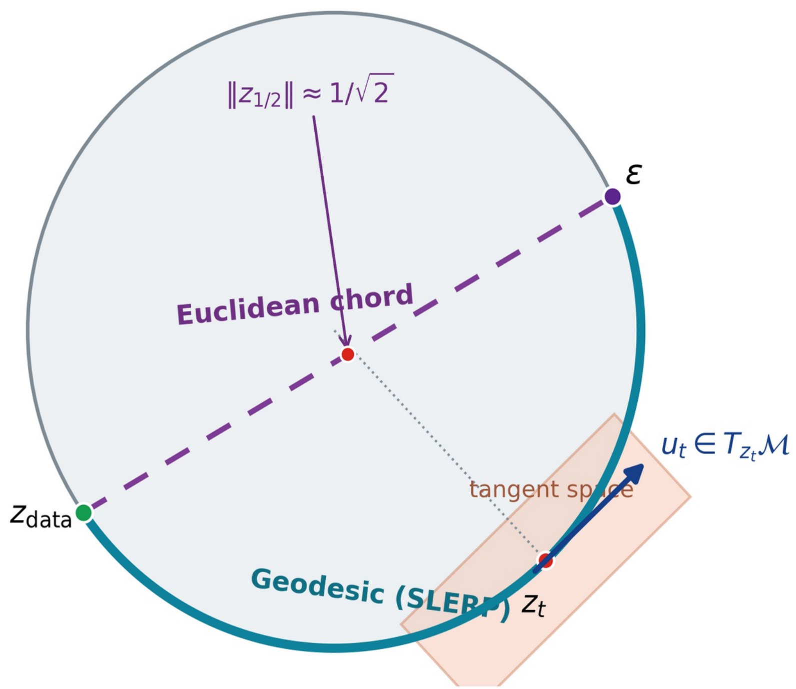
$$z_{t-\Delta t} = \mathrm{Exp}_{z_t}\!\left(-\Delta t\, \Pi_{z_t}(v_\theta)\right)$$

The representation, training path, and numerical update use the same geometry.

Why the autoencoder and prior must match

Two controlled tests separate path mismatch from support mismatch

Same spherical endpoints, different transport path



1. Path match: same spherical autoencoder

Method	FID	KID	Prec.	Recall	Min norm
Chord	67.44	0.0740	0.840	0.085	0.705
SRUL	**63.84**	**0.0667**	**0.857**	**0.091**	**1.000**

2. Support match: SRUL vs. the same VAE with two priors (CFG = 2)

Method	FID	Prec.	Recall
SRUL: spherical AE + RFM	**49.42**	**0.857**	**0.110**
VAE + standard FM	50.70	0.855	0.091
VAE + projected RFM	56.04	0.848	0.063

Same VAE: radius CV = 0.291; mean-radius replacement changes reconstruction FID 16.90 → 22.46.

SRUL learns spherical support before RFM. Projecting a VAE after training removes radius information.

Information control and conditional sampling

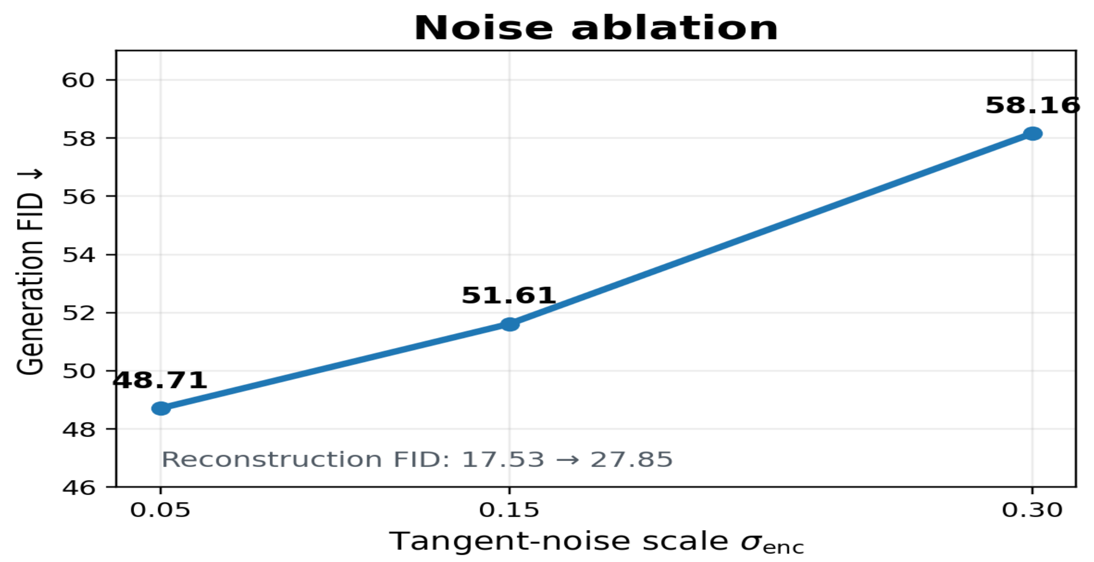
σenc controls retained image information; CFG controls class emphasis during sampling

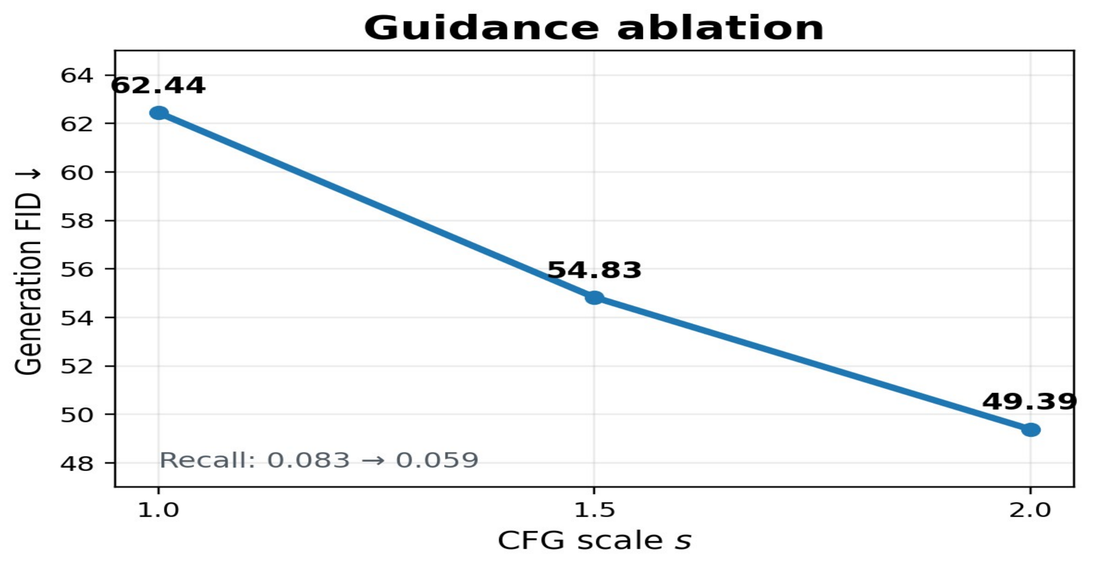
Tangent-noise information control

$$z_\sigma = \mathrm{Exp}_z(\sigma_{\mathrm{enc}}\,\Pi_z(\xi))$$

Classifier-free guidance

$$v_{\mathrm{CFG}} = v_{\mathrm{uncond}} + s\,(v_{\mathrm{cond}} - v_{\mathrm{uncond}})$$



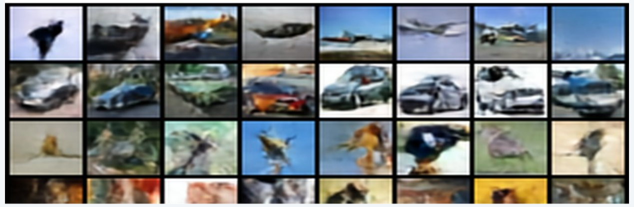


σenc = 0.05 gives the best tested balance; larger noise removes too much image information.

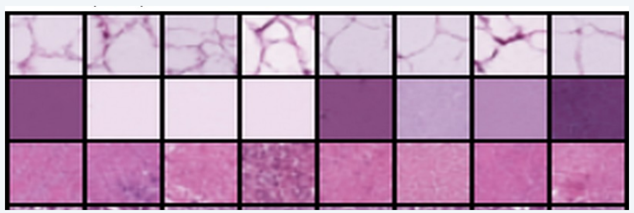
Larger s improves FID and precision, but concentrates samples and lowers recall.

SRUL in three image settings

The same method is tested on different image structure and resolution



CIFAR-10



PathMNIST

CelebA-64

Dataset	Evaluation setting	Grid	Recon. FID	Gen. FID	Prec.	Recall
CIFAR-10	Geometry + sampling ablations	4 × 4	17.53	48.71	0.861	0.090
PathMNIST	Histopathology transfer	4 × 4	7.13	29.36	0.829	0.309
CelebA-64	64 × 64 face generation	8 × 8	9.08	41.22	0.657	0.198

CIFAR-10 tests geometry and information control; PathMNIST tests texture transfer; CelebA-64 tests 64 × 64 faces.